

第 1 周：资源清算、环境搭建与 PCA-B0 定义

详细实验 runbook：每一步都有命令、输出和验收标准。

总原则：先搭骨架，再跑实验

这一版不是“计划表”，而是实验室操作手册。每一步都有输入、命令、输出和验收标准。第 1 周禁止训练模型，禁止下载所有大数据，禁止开始写论文结论。

本周最终产物	必须满足的条件
目录骨架	PCA-B0 项目目录可复制，README 有目标和边界。
环境 smoke test	torch、diffusers、transformers、datasets 至少能 import。
资源矩阵	每个资源登记 paper/repo/data/weights/license/是否第一阶段使用。
Schema	pca_b0_schema.json 包含 task、candidate、score、human eval 字段。
Week2 就绪	知道先跑 ImageReward、EmotionCLIP、CLIP subject、GenEval 简化版。

Day 1 - 项目目录和 README

目标：创建不会混乱的项目目录。后续所有数据、图像、分数、报告都必须写到固定位置。

```
cd ~/projects
mkdir -p PCA-B0
cd PCA-B0
mkdir -p data/{raw,external,processed,pilot,personal,cache,references}
mkdir -p outputs/{candidates,scores,ranked,figures,logs,contact_sheets}
mkdir -p scripts/{scorers,generation,pipeline,analysis,human_eval}
mkdir -p configs reports docs notebooks resources/{papers,repos,links,licenses}
touch README_research.md experiment_log.md resource_matrix.csv
```

验收：执行 `find . -maxdepth 2 -type d | sort`，能看到上述目录。README_research.md

至少写明：目标、非目标、当前阶段产物。

```
cat > README_research.md <<'MD'
# PCA-B0: Perceptual-Consistency Activation Baseline
```

Goal: first filter candidates by core consistency, then rank accepted candidates by perceptual activation.

Non-goals in phase 1:

- no foundation-model training
- no claim that emotion score equals user memory activation
- no claim that subject consistency equals pixel copying

Phase 1 deliverables:

- scorer adapters
- pilot task set
- gate/rank pipeline
- small human-evaluation protocol

MD

Day 2 - Python/Conda 环境

目标：保证后续脚本能运行。先装通用依赖；特殊仓库例如 GenEval/RelTR 之后单独装。

```
conda create -n pca-b0 python=3.10 -y
conda activate pca-b0
python -m pip install --upgrade pip
cat > requirements.txt <<'REQ'
torch
torchvision
torchaudio
transformers>=4.40
diffusers>=0.27
accelerate
safetensors
datasets
huggingface_hub
Pillow
opencv-python
pandas
numpy
scikit-learn
tqdm
pyyaml
matplotlib
open_clip_torch
image-reward
easyocr
REQ
pip install -r requirements.txt
```

常见失败处理：如果 torch 装错 CUDA，不要当天死磕；先用 CPU 跑资源登记和评分器导入，生成图像放到 Week3。

Day 3 - Hugging Face 登录、缓存和权限

目标：所有模型和数据下载都进固定缓存目录，不把模型写满系统盘。

```
mkdir -p ~/.hf_cache
export HF_HOME=~/.hf_cache
export HF_HUB_CACHE=~/.hf_cache/hub
huggingface-cli login
python - <<'PY'
from huggingface_hub import whoami
try: print(whoami())
except Exception as e: print('HF login not ready:', e)
PY
```

记录到 experiment_log.md：机器名、GPU 型号、CUDA、Python 版本、HF_HOME 路径。

```
python - <<'PY'
import torch, sys
print('python', sys.version)
print('torch', torch.__version__)
print('cuda available', torch.cuda.is_available())
if torch.cuda.is_available(): print(torch.cuda.get_device_name(0))
PY | tee outputs/logs/env_check.txt
```

Day 4 - 资源矩阵逐项清算

目标：不是读完所有论文，而是确定哪些资源第一阶段可运行、哪些只是理论支撑、哪些需要许可证或推迟。打开每个链接后只记录6个字段：数据、代码、权重、输入、输出、风险。

资源	第一周怎么处理	下载/打开链接
SDXL Base 1.0	Week2/3 会安装或调用；先登记。	https://huggingface.co/stabilityai/stable-diffusion-xl-base-1.0
Diffusers SDXL docs	Week2/3 会安装或调用；先登记。	https://huggingface.co/docs/diffusers/en/using-diffusers/sdxl
PartiPrompts	可小样本下载；用于 pilot。	https://huggingface.co/datasets/nateraw/parti-prompts
DiffusionDB	登记链接，先不下载全量。	https://huggingface.co/datasets/poloclub/diffusiondb
EmoSet GitHub	登记链接，先不下载全量。	https://github.com/JingyuanYY/EmoSet
EmoSet project	登记链接，先不下载全量。	https://vcc.tech/EmoSet
ArtEmis GitHub	登记链接，先不下载全量。	https://github.com/optas/artemis
ArtEmis paper	登记链接，先不下载全量。	https://openaccess.thecvf.com/content/CVPR2021/html/Achlioptas_ArtEmis_Affective_Language_for_Visual_Art_CVPR_2021_paper.html
EmotionCLIP GitHub	Week2/3 会安装或调用；先登记。	https://github.com/Xeaver/EmotionCLIP
EmotionCLIP HF	Week2/3 会安装或调用；先登记。	https://huggingface.co/jiangchengchengNLP/EmotionCLIP
LaMem/MemNet	登记链接，先不下载全量。	https://memorability.csail.mit.edu/download.html
PickScore GitHub	Week2/3 会安装或调用；先登记。	https://github.com/yuvalkirstain/pickscore
PickScore HF	Week2/3 会安装或调用；先登记。	https://huggingface.co/yuvalkirstain/PickScore_v1
ImageReward GitHub	Week2/3 会安装或调用；先登记。	https://github.com/zai-org/ImageReward
ImageReward HF	Week2/3 会安装或调用；先登记。	https://huggingface.co/zai-org/ImageReward

资源	第一周怎么处理	下载/打开链接
HPSv2 GitHub	Week2/3 会安装或调用；先登记。	https://github.com/tgxs002/HPSv2
DreamBench++ GitHub	可小样本下载；用于 pilot。	https://github.com/yuangpeng/dreambench_plus
DreamBench++ project	可小样本下载；用于 pilot。	https://dreambenchplus.github.io/
VBench GitHub	登记链接，先不下载全量。	https://github.com/Vchitect/VBench
VBench project	登记链接，先不下载全量。	https://vchitect.github.io/VBench-project/
T2I-CompBench GitHub	可小样本下载；用于 pilot。	https://github.com/Karine-Huang/T2I-CompBench
T2I-CompBench project	可小样本下载；用于 pilot。	https://karine-h.github.io/T2I-CompBench/
GenEval GitHub	可小样本下载；用于 pilot。	https://github.com/djghosh13/geneval
RelTR GitHub	登记链接，先不下载全量。	https://github.com/yrcong/RelTR
GeoDE GitHub	登记链接，先不下载全量。	https://github.com/princetonvisualai/geode_dataset
GeoDE paper	登记链接，先不下载全量。	https://arxiv.org/html/2301.02560v4
Dollar Street	登记链接，先不下载全量。	https://www.gapminder.org/dollar-street
Dollar Street paper	登记链接，先不下载全量。	https://openreview.net/forum?id=qnfYsave0U4
DIG-In GitHub	登记链接，先不下载全量。	https://github.com/facebookresearch/DIG-In
DIG-In paper	登记链接，先不下载全量。	https://arxiv.org/html/2308.06198v3
THINGS	登记链接，先不下载全量。	https://things-initiative.org/
THINGS paper	登记链接，先不下载全量。	https://journals.plos.org/plosone/article?id=10.1371%2Fjournal.pone.0223792

```
cat > resource_matrix.csv <<'CSV'
```

```
resource_name,layer,paper_url,repo_url,dataset_url,weights_url,license,phase1_use,download_now,risk,notes
```

```
ImageReward,preference,https://arxiv.org/abs/2304.05977,https://github.com/zai-org/ImageReward,,https://huggingface.co/zai-org/ImageReward,check,yes,yes,low,primary  
preference scorer
```

```
EmotionCLIP,emotion,https://arxiv.org/abs/2305.13500,https://github.com/Xeaver/EmotionCLIP,,https://huggingface.co/jiangchengchengNLP/EmotionCLIP,check,yes,yes,medium,  
scorer; fallback to CLIP zero-shot
```

```
GenEval,structure,https://papers.nips.cc/paper\_files/paper/2023/hash/a3bf71c7c63f0c3bcb7ff67c67b1e7b1-Abstract-Datasets\_and\_Benchmarks.html,https://github.com/djghosh13/gen  
evaluator
```

```
CSV
```

Day 5 - Smoke tests：不生成图，只验证库能用

```
cat > scripts/pca_b0_smoke_test.py <<'PY'
import importlib
mods = ['torch','transformers','diffusers','datasets','PIL','pandas','numpy','open_clip']
for m in mods:
    try:
        mod = importlib.import_module(m)
        print('[OK]', m, getattr(mod, '__version__', 'version unknown'))
    except Exception as e:
        print('[FAIL]', m, e)
print('Smoke test finished.')
PY
python scripts/pca_b0_smoke_test.py | tee outputs/logs/week1_smoke_test.txt
```

再验证 Hugging Face datasets 能读取小数据。

```
python - <<'PY'
from datasets import load_dataset
ds = load_dataset('nateraw/parti-prompts', split='train')
print(ds[0])
print('rows', len(ds))
PY
```

Day 6 - Schema 和 Pilot Task 模板

目标：后续所有候选图、分数、人类标注都进入同一张表。不要用散乱文件名推断实验。

```
{
  "task_id": "T001",
  "prompt": "text prompt",
  "reference_image": "path/url optional",
  "user_preference_statement": "what user values",
  "core_value_tags": [
    "friendly",
    "rounded"
  ],
  "forbidden_loss": [
    "cold",
    "aggressive"
  ],
  "allowed_deviation": [
    "color",
    "pose"
  ],
  "task_type": "subject|spatial|count|text|culture|emotion",
  "target_region": "optional",
  "expected_text": "optional",
  "candidate_id": "T001_raw_s1000",
  "image_path": "outputs/candidates/...png",
  "intervention_level": "raw|retry|rewrite|matched|mismatched|oracle",
  "seed": 1000,
  "model_id": "stabilityai/stable-diffusion-xl-base-1.0",
  "subject_gate_score": null,
  "structure_gate_score": null,
  "accepted_by_gate": null,
  "imagereward": null,
  "hpsv2": null,
  "emotion_top1": null,
  "emotion_conf": null,
  "memorability_score": null,
  "culture_score": null,
  "activation_rank_score": null,
  "rank_in_task": null,
  "human_choice": null,
  "human_reason": "optional"
}
```

创建 3 条最小任务。

```
cat > data/pilot/tasks_smoke.csv <<'CSV'
task_id,prompt,reference_image,user_preference_statement,core_value_tags,forbidden_loss,allowed_deviation,task_type,target_region,expected_text
T001,"a friendly taxi driver mascot, warm and approachable",data/references/taxi_ref.png,"keep friendly rounded city-life feeling","friendly;rounded;urban","cold;aggressive;robotic","pose;color;clothing",subject,,
T002,"a red cube to the left of a blue sphere",,"red cube;blue sphere;left-of","wrong relation","style;background",spatial,,
T003,"a sign that says LOCAL",,"text LOCAL","misspelled text","font;background",text,,LOCAL
CSV
```

Day 7 - 周报告和进入 Week2 的 Go/No-Go

如果以下 5 项没完成，不进入 Week2：环境 smoke test、资源矩阵、schema、3 条 smoke tasks、Week2 scorer 优先级。

检查项	通过标准
环境	outputs/logs/week1_smoke_test.txt 中关键库 OK。
资源	resource_matrix.csv 至少填 15 个资源。
Schema	common/pca_b0_schema.json 或项目内 pca_b0_schema.json 存在。
Pilot	data/pilot/tasks_smoke.csv 存在并能被 pandas 读取。
风险	reports/week1_report.md 写明卡点和推迟项。